

RippleWealth: AI-Driven Investment Risk Intelligence Platform

Sprint 3 Deliverable:

Computer Science Capstone Project CS 691

Prof. Kshitij Sharma • Spring 2026

Group 8:

Sharvani Kadarla • Deepali Sawant • Jai Sheel Bhasker Paladugu

• Ashish Reddy Majjigapu



Agenda

- Team Member Roles & Responsibilities
- Improvements from Professor Feedback
- Problem Statement
- Project Description
- Team Working Agreement
- Personas
- Minimum Viable Product (MVP)
- System Design & Technologies
- Algorithms Used
- System Diagrams
- Sprint 2 Recap
- Product Backlog
- Sprint Summary (Sprint 0, 1, 2)
- Sprint 3 Backlog
- Test Cases (Sprint 3)
- Sprint Metrics
- Sprint Retrospective
- Project Demonstration
- GitHub Repository & Technical Paper
- Live Application

Team Member Roles and Responsibilities

Our project success hinges on a clearly defined structure, where each team member's expertise drives specific outcomes.



Sharvani Kadarla

Project Manager & Software Developer

- Define project vision, roadmap, and milestones
- Plan sprints and coordinate team tasks
- Design system architecture and data flow
- Ensure integration across all modules
- Monitor progress and resolve blockers



Deepali Sawant

Frontend & Visualization Engineer

- Develop interactive UI and dashboards
- Build portfolio and knowledge graph visualizations
- Ensure clean, user-friendly UI/UX design
- Integrate frontend with backend APIs and AI outputs



Jai Sheel Bhasker Paladugu

AI / Machine Learning Engineer

- Develop AI models for event impact analysis
- Build causal graphs for industry and company effects
- Implement LLM-based explainable insights
- Run scenario simulations for portfolio risk analysis



Ashish Reddy Majjigapu

Backend & Data Engineering Lead

- Develop RESTful APIs using FastAPI
- Build data ingestion and preprocessing pipelines
- Manage data storage and validation
- Support portfolio risk calculations and backend performance

Improvements Made From Professor's Feedback

After reviewing the professor's feedback on our sprint evaluation, we identified the need to improve planning, prioritization, and timely task completion.

Better Sprint Planning

Refined user stories and personas to better reflect target user needs, enabling clearer task prioritization and more effective assignment of responsibilities early in the sprint cycle.

Improved Scheduling & Retrospectives

Optimized sprint timelines and task distribution for consistency across deliverables, and strengthened retrospectives by capturing key insights, challenges, and improvements to support continuous agile progress.

Progress Tracking

Introduced clear internal deadlines and improved milestone tracking to ensure timely progress and prevent delays during Sprint 3 execution.

Enhanced Communication

Strengthened communication through regular meetings and status updates, resolving blockers quickly

These improvements helped the team maintain better alignment, accountability, and more efficient sprint execution for the RippleWealth project.

Problem Statement

Reactive, Not Proactive

Modern investment platforms alert investors only after stock prices change, missing early warning signals

Ignored Risk Triggers

Current tools focus mainly on financial indicators and ignore non-financial risk triggers like supply chain disruptions, weather events, labor strikes, or logistics delays

Lack of Clear Explanations

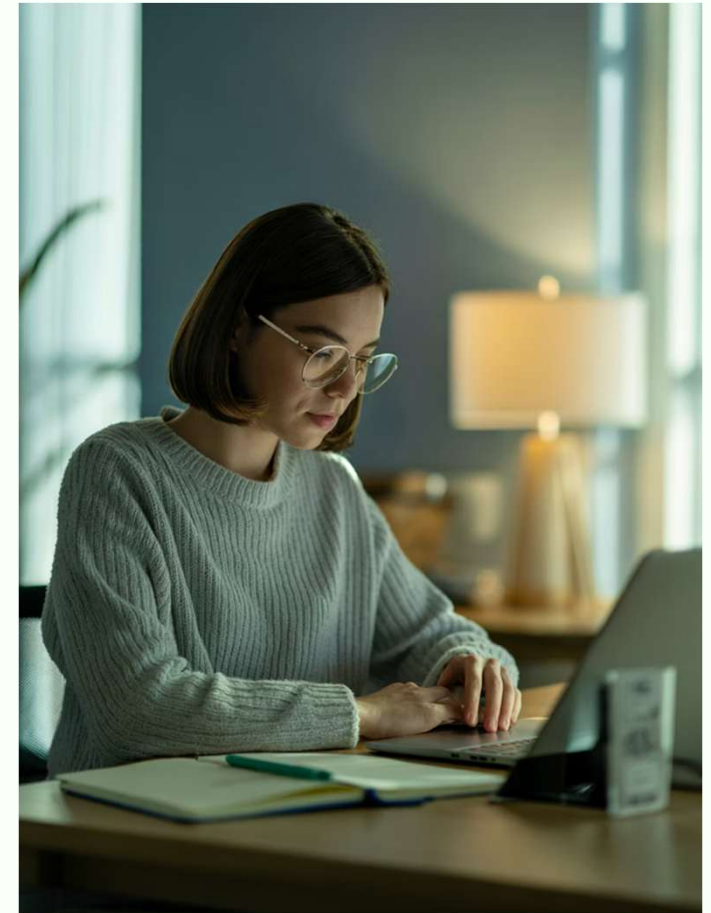
Investors lack clear explanations of why their portfolio risk is increasing and cannot anticipate risk or make proactive portfolio decisions



Project Description

RippleWealth is an AI-driven investment risk intelligence platform that anticipates market risks by detecting real-world disruptions—such as weather events, supply chain delays, and labor strikes—before they appear in stock prices. Unlike traditional investment tools that react only after price changes, RippleWealth provides early insights to help investors understand potential portfolio impacts.

Using a multi-agent AI architecture and LLM-based causal reasoning, the platform maps how these events propagate across industries and impact companies and portfolios. Through an interactive Knowledge Graph and explainable risk metrics like Value-at-Risk (VaR), RippleWealth enables proactive, transparent, and data-driven portfolio risk management.



Team Working Agreement

Participation

- Attend scheduled meetings on time
- Come prepared with progress updates
- Inform the team leader in advance if absent

Meetings

- Weekly meetings via Zoom / Google Meet
- Team leader schedules and shares meeting links
- Meeting minutes recorded after each session

Work Quality & Collaboration

- Share work early for feedback
- Maintain professionalism and accountability
- Ensure alignment with project goals

Work Division & Ownership

- Tasks assigned during sprint planning
- Equal responsibility among team members
- Meet agreed deadlines and communicate blockers early

Communication Tools

- Zoom / Google Meet - Sprint meetings
- WhatsApp - Quick updates and urgent communication
- Google Docs & Slides - Collaborative deliverables

Team Transparency

- Cameras encouraged during meetings
- Maintain open communication and team trust

User Persona 1: Ethan Miller

Young Professional & Retail Investor



Age: 29 **Location:** Austin, Texas **Job:** Software Engineer **Salary:** \$95K - \$120K/year
Family: Lives with partner **Investment Level:** Beginner to Intermediate

Goals

- Understand Value-at-Risk clearly
- Simulate market crash scenarios
- Get simple portfolio risk scores
- Improve long-term returns
- Make safer investment decisions

Frustrations

- Cannot clearly measure portfolio risk
- Hard to understand downside scenarios
- Too many scattered finance tools
- Confusing probability models
- Lacks simple visual risk insights

Interests

Personal finance and investing • Portfolio tracking apps • Market trend analysis
• Tech and automation tools • Financial podcasts • Long-term wealth building

User Persona 2: Sophia Chen

Financial Risk Analyst (Professional)



Age: 38 **Location:** New York, NY **Job:** Financial Risk Analyst **Salary:** \$130K - \$160K/year **Family:** Married, one child **Investment Level:** Advanced / Professional

Goals

- Run large-scale Monte Carlo simulations
- Identify systemic risk pathways
- Model inter-company relationships
- Generate client risk reports quickly
- Improve predictive accuracy

Frustrations

- Manual scenario modeling is time-consuming
- Tools lack integrated causal analysis
- Hard to map indirect market impacts
- Needs faster simulation turnaround
- Limited explainability in some models

Interests

Risk modeling techniques • Market stress testing • Portfolio optimization • Quantitative finance • Economic indicators • Financial data visualization

User Persona 3: Carlos Ramirez

Business Owner & Wealth Planner



Age: 45 **Location:** Atlanta, Georgia **Job:** Small Business Owner **Salary:** \$110K - \$150K/year
Family: Married **Investment Level:** Moderate Investor

Goals

- Understand event → market → portfolio impact
- Visualize causal financial relationships
- Stress-test business investments
- Reduce exposure to high-risk sectors
- Make resilient investment plans

Frustrations

- Unsure how world events affect investments
- No clear impact chain visibility
- Market shocks feel unpredictable
- Hard to evaluate sector dependencies
- Lacks scenario-based planning tools

Interests

Business growth strategy • Market and sector trends • Supply chain news • Financial planning tools • Diversified investments • Economic forecasts

Minimal Viable Product (MVP)

Alternative Data Engine

Modern investment platforms often alert investors only after stock prices change, which can cause early warning signals to be missed.

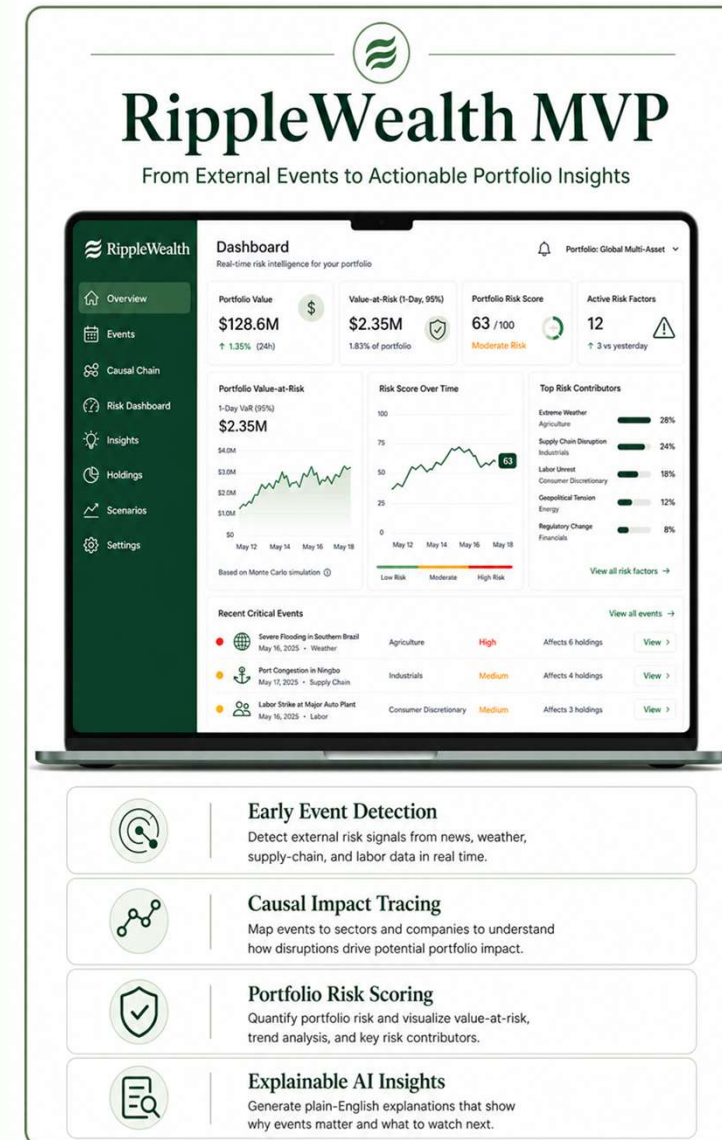
Ripple Wealth detects external risk signals from news, weather, supply-chain disruptions, and labor events using keyword filtering and NLP.

Portfolio Risk Dashboard

Displays Value-at-Risk, portfolio risk scores, and scenario-based analysis using Monte Carlo simulation.

Explainable Insights

Generates plain-English explanations showing how detected events influence holdings and overall portfolio risk.



Core Technologies Powering Our Platform

AI / Analytics

Python, Monte Carlo simulation, Value-at-Risk, and keyword NLP support early-event detection and portfolio risk modeling.

Backend APIs

FastAPI powers REST endpoints for portfolio data, event processing, risk scoring, and frontend integration.

Frontend

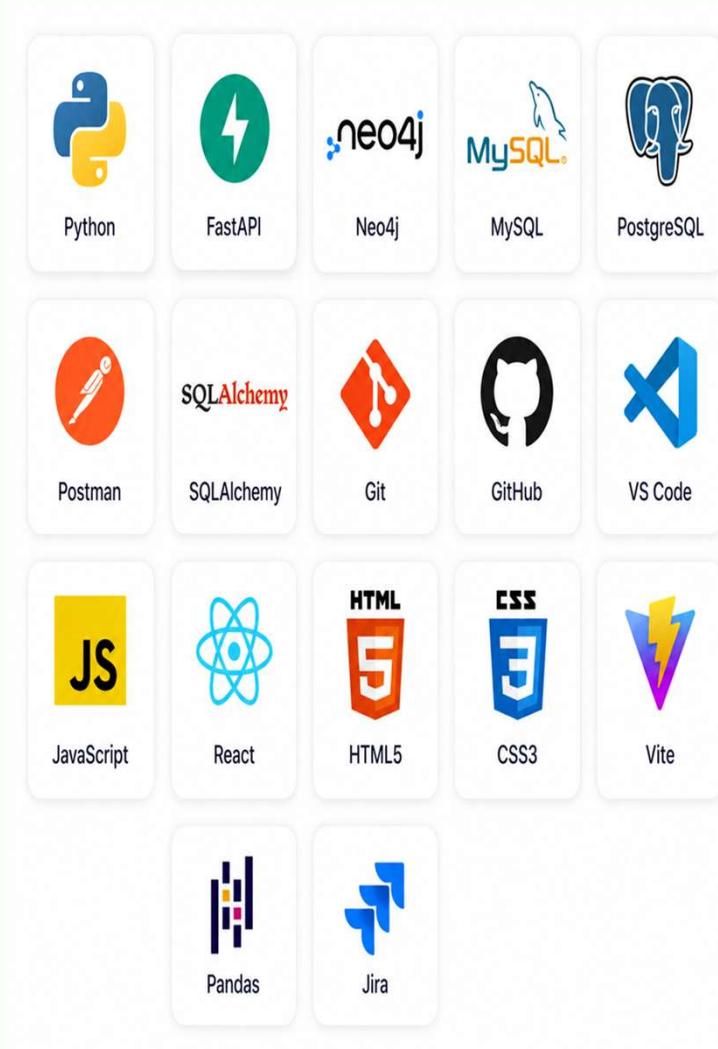
React.js, JavaScript, HTML5, and CSS3 support the dashboard, interaction flow, and visual components.

Graph Technology

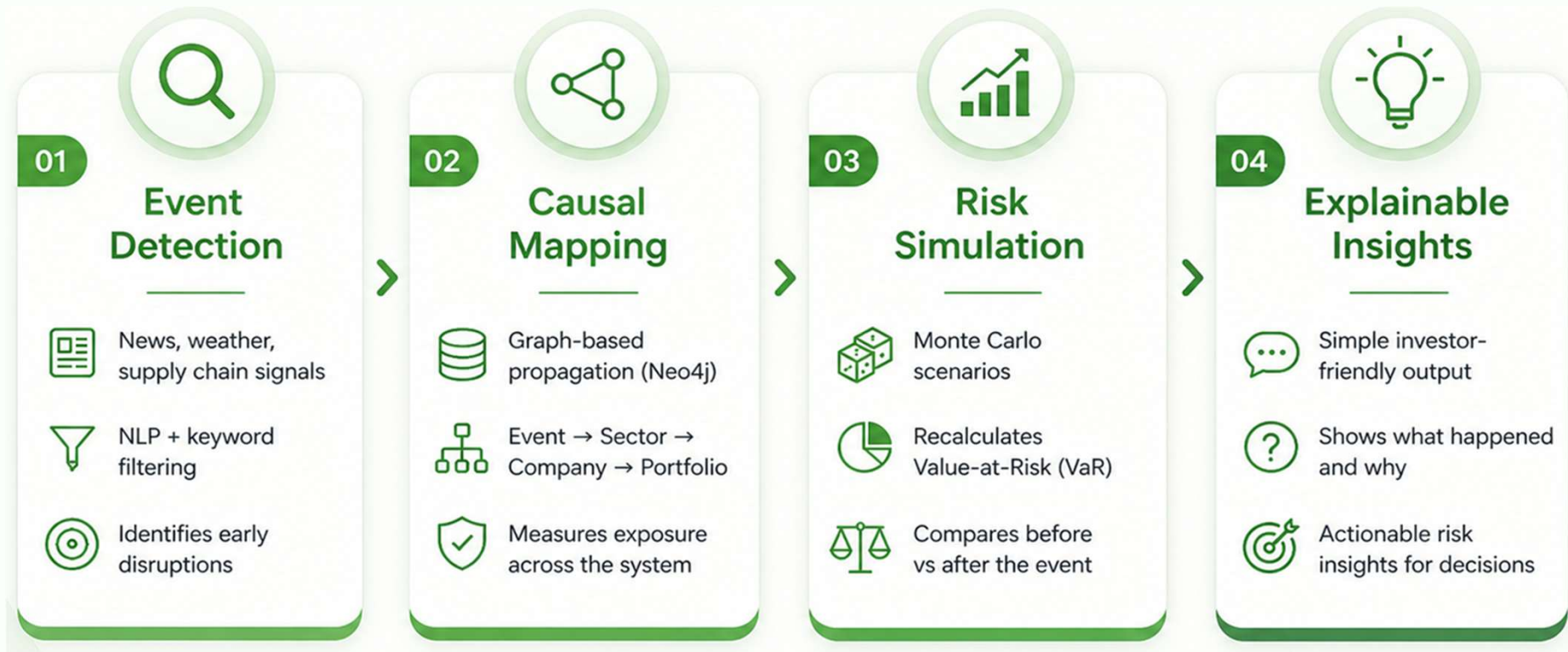
Neo4j supports impact tracing and causal relationship visualization across sectors, companies, and holdings.

Development Tools

Git, GitHub, Postman, VS Code, and Jira support version control, API testing, development, and sprint management.

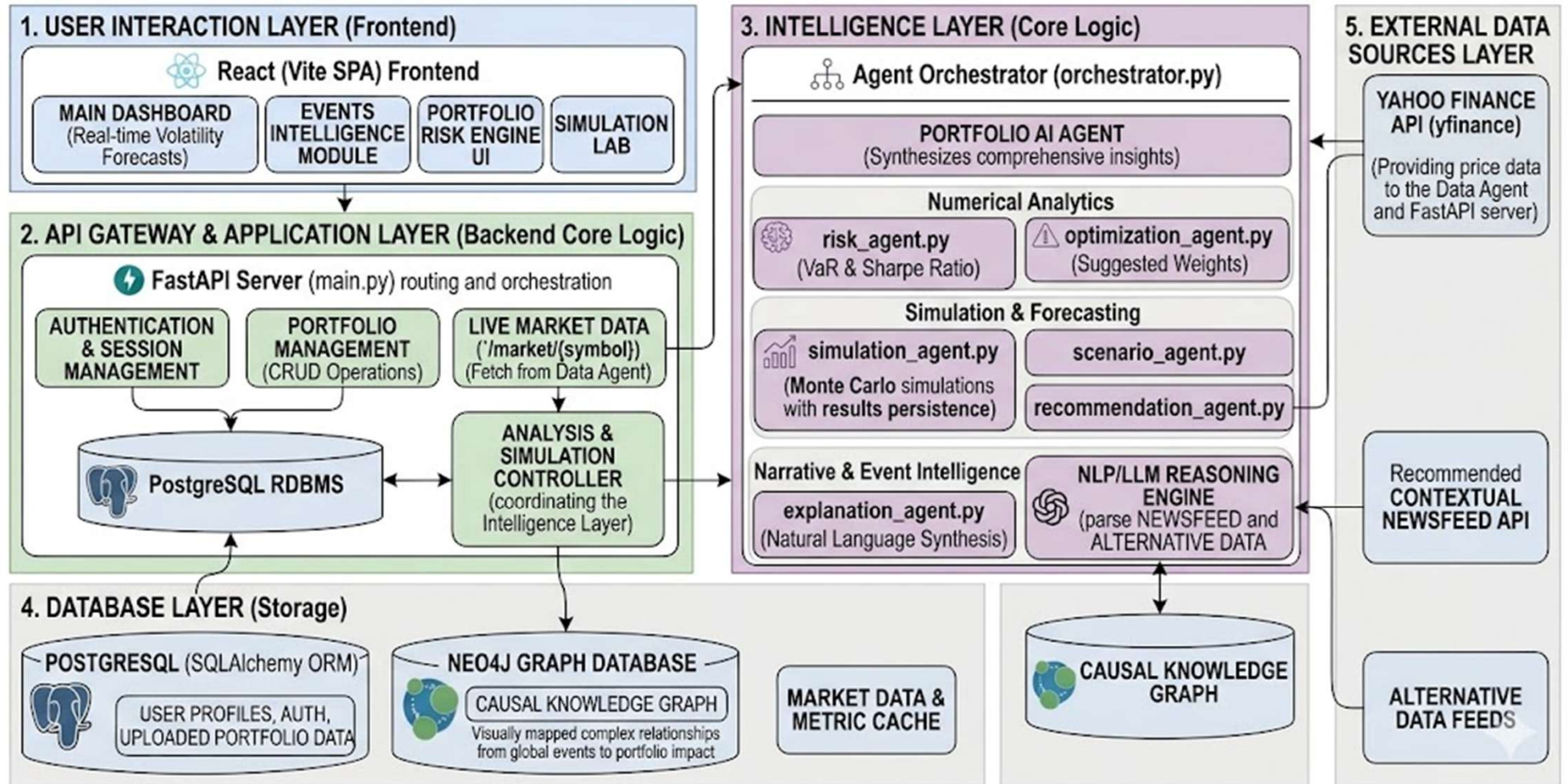


Algorithms

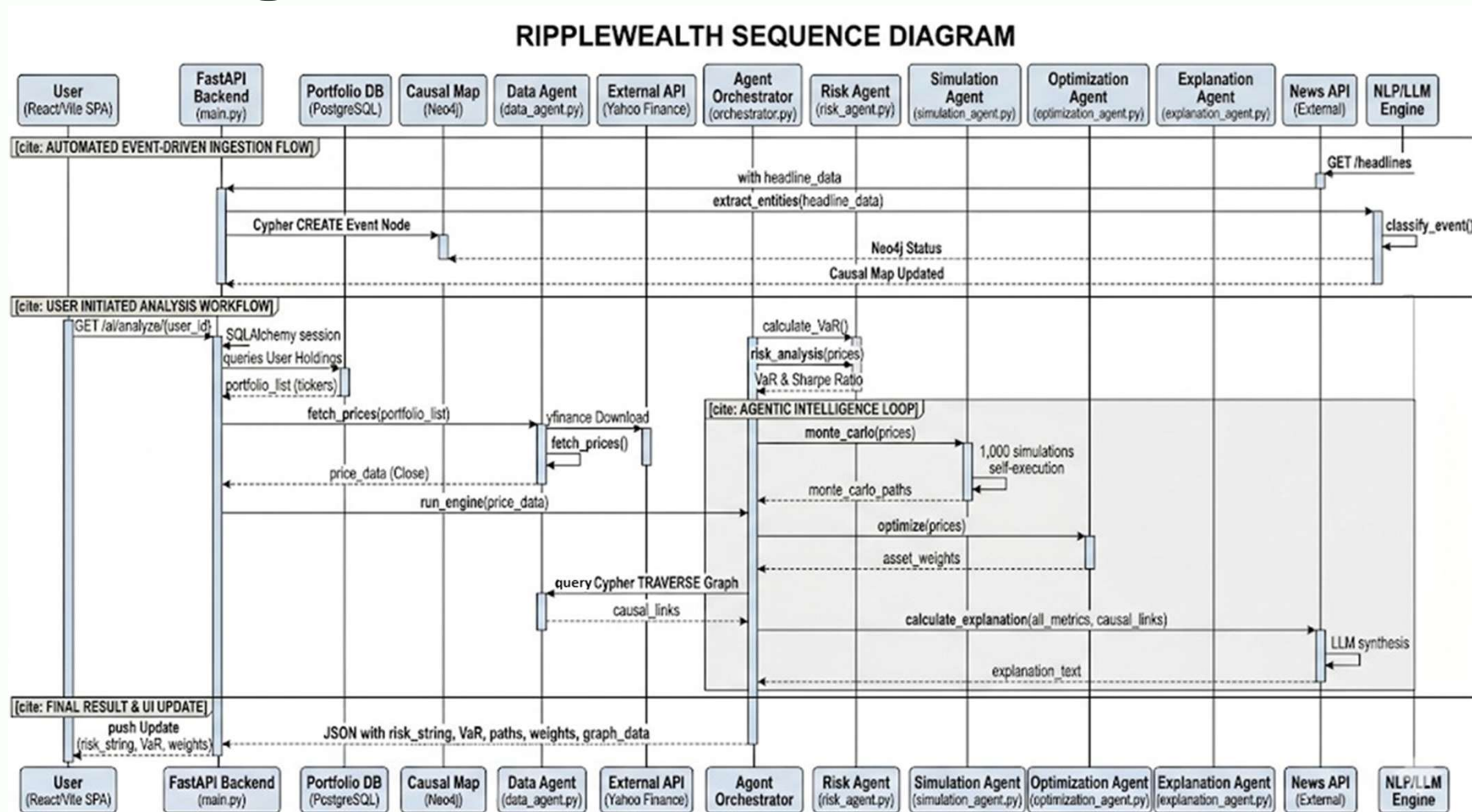


System Architecture Diagram

RIPPLEWEALTH SYSTEM ARCHITECTURE DIAGRAM

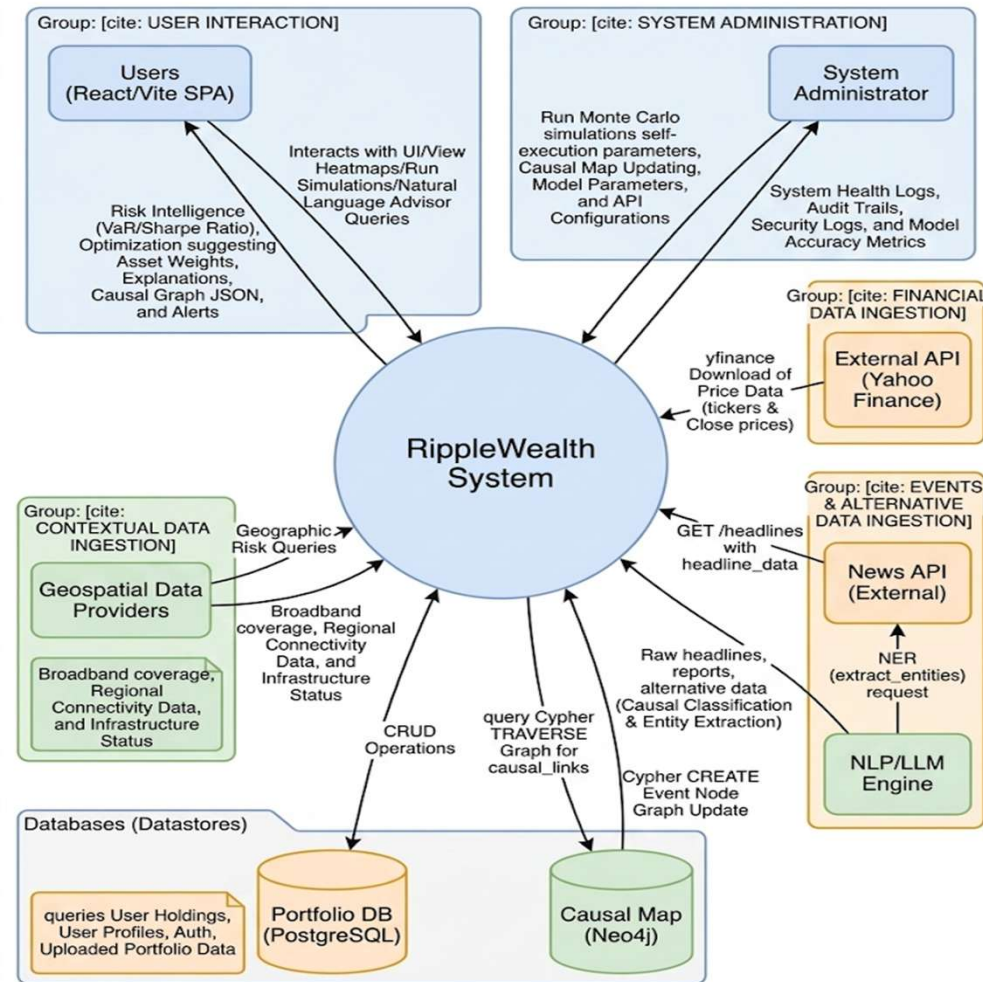


Sequence Diagram



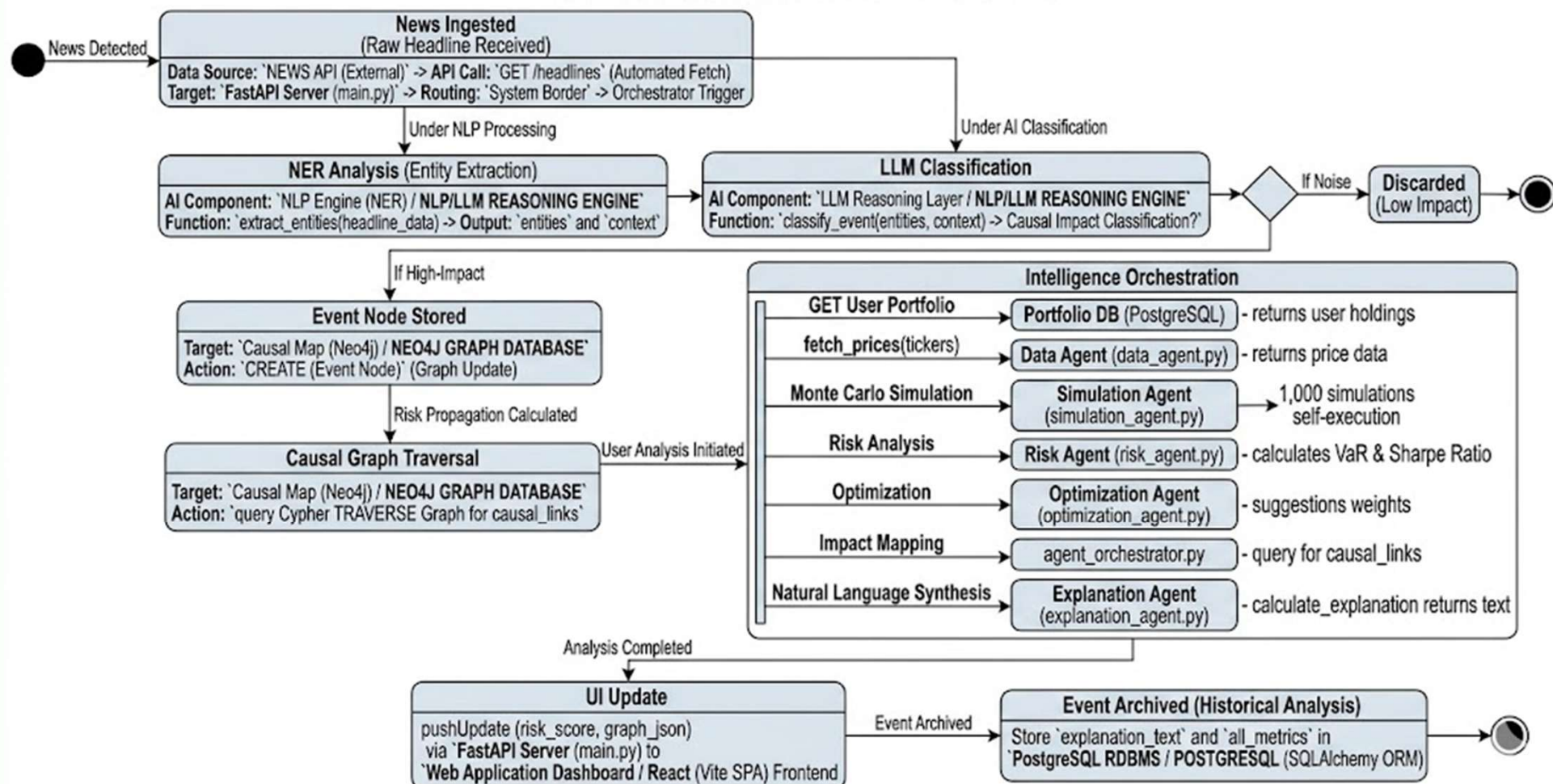
Context Diagram

RIPPLEWEALTH CONTEXT DIAGRAM



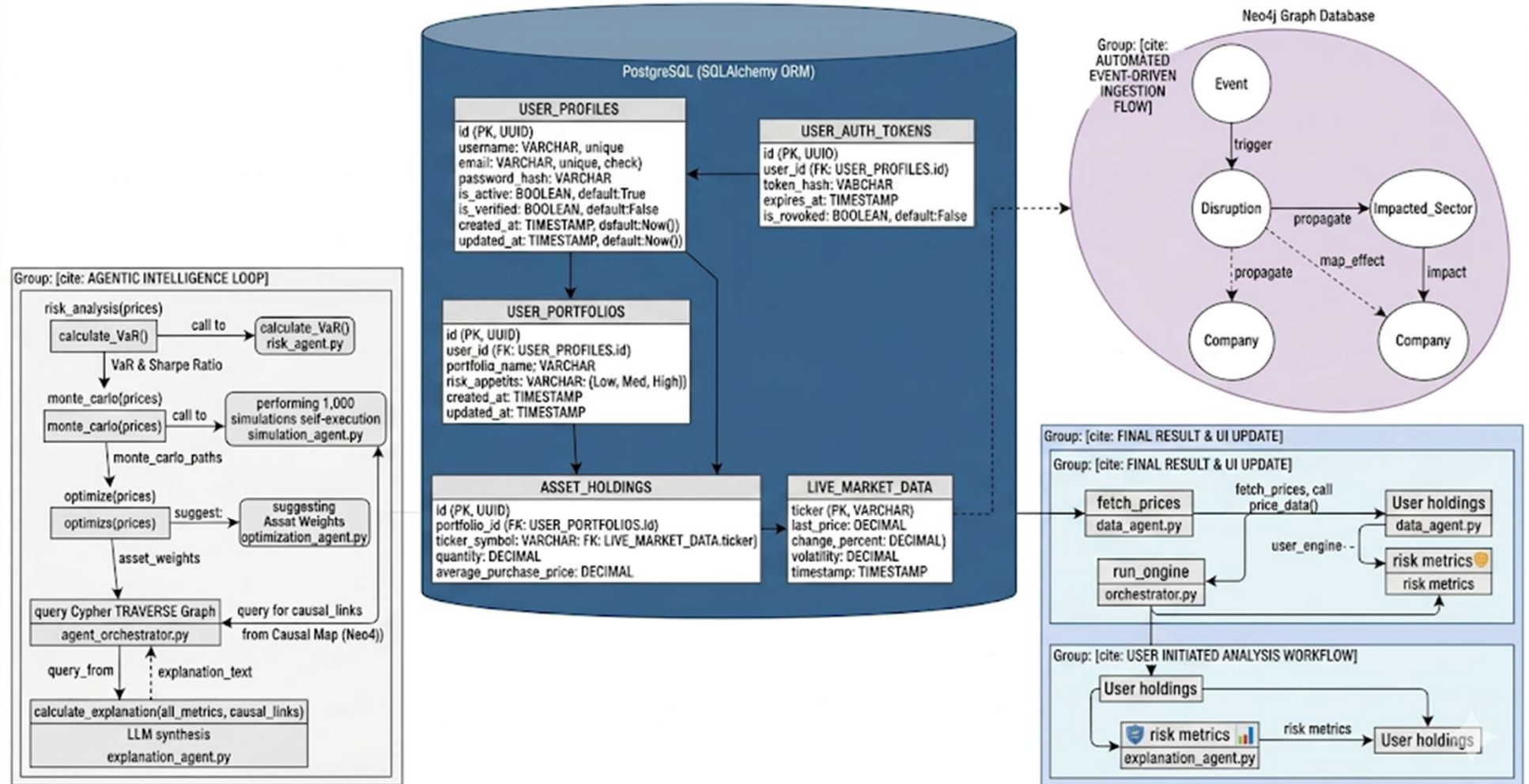
State Diagram

RIPPLEWEALTH STATE DIAGRAM



Database Diagram

RIPPLEWEALTH DATABASE SCHEMA DIAGRAM



Sprint 2 Recap

Sprint Goal:

Deliver a fully integrated RippleWealth platform with responsive UI, live event-driven simulation, end-to-end API connectivity

Work Completed:

- Setup FastAPI Project & Database Models
- Knowledge Graph Visualization Structure
- AI- Backend Integration

Key Features:

- End-to-end authentication – register, login, and persistent session across page refreshes
- Centralised risk dashboard with sector allocation bars, KPI cards, and metric cards
- Error handling
- Natural language risk explanations generated for every analysis run

Product Backlog - User Stories

☐ Team-8 Sprint 3 8 Apr – 5 May (14 work items)	26533	Complete sprint	...
Finish Development of Frontend, Finish Development of Backend, Frontend–Backend Integration Working, System Testing, Finish Project Demo, Finish Technical Paper, Work on Deliverable 4			
T8-70 As an investor, I want to complete the full workflow from portfolio upload to risk insights, so that I can evaluate the pl...	TO DO ▾	8	
T8-71 As a user, I want secure session handling (login/logout/session timeout), so that my account remains protected.	TO DO ▾	5	
T8-72 As a user, I want the application to work across different browsers, so that I can access it from any environment.	TO DO ▾	3	
T8-73 As an investor, I want event data to refresh dynamically, so that I stay updated on global disruptions.	TO DO ▾	3	
T8-74 As a user, I want fast response times, so that I can interact efficiently.	TO DO ▾	2	
T8-75 As a user, I want to ask basic questions about portfolio risk, so that I can get quick insights.	TO DO ▾	3	
T8-76 As a user, I want quick access buttons on the home page for direct navigation.	TO DO ▾	2	

Acceptance Criteria

Issue Type	ID	Summary	Scenario
Acceptance Criteria	T8-70	As an investor, I want to complete the full workflow from portfolio upload to risk insights, so that I can evaluate the platform end-to-end.	Given I am a logged-in investor When I upload a portfolio file and navigate through all workflow steps Then I receive complete risk insights without errors
Acceptance Criteria	T8-71	As a user, I want secure session handling (login/logout/session timeout), so that my account remains protected.	Given I am a registered user When I log in, remain idle, or explicitly log out Then my session is protected and terminated appropriately
Acceptance Criteria	T8-72	As a user, I want the application to work across different browsers, so that I can access it from any environment.	Given I am using a supported browser When I access the application Then all features render and function correctly
Acceptance Criteria	T8-73	As an investor, I want event data to refresh dynamically, so that I stay updated on global disruptions.	Given I am viewing the disruption events dashboard When new event data becomes available Then the dashboard updates without requiring a manual page refresh
Acceptance Criteria	T8-74	As a user, I want fast response times, so that I can interact efficiently.	Given I am interacting with any part of the application When I perform an action (navigation, search, load) Then the system responds within acceptable time thresholds
Acceptance Criteria	T8-75	As a user, I want to ask basic questions about portfolio risk, so that I can get quick insights.	Given I am on the portfolio agent page and my portfolio data is available When I enter a question about portfolio risk (e.g., "What is my risk exposure?") Then I receive a relevant response explaining the portfolio risk based on available data
Acceptance Criteria	T8-76	As a user, I want quick access buttons on the home page for direct navigation.	Given I am on the home page When I view the quick access section Then I can navigate directly to key features with a single click

Test Cases

Test Case ID	Module	Test Objective	Test Steps	Expected Result
TC01	Authentication	Verify user login functionality	1. Launch the application 2. Navigate to Login page 3. Enter valid credentials 4. Click Login	User is successfully logged in and redirected to the dashboard
TC02	Authentication	Verify logout functionality	1. Log in to the application 2. Click Logout button	User is logged out and session is terminated securely
TC03	Portfolio Workflow	Verify full workflow from adding tickers to risk insights	1. Log in as investor 2. Add tickers 3. Navigate through all workflow steps 4. View risk insights	Complete risk insights are generated without errors
TC04	Events	Verify events dashboard auto-refreshes dynamically	1. Navigate to Events dashboard 2. Look for live events	Events dashboard updates automatically without manual refresh
TC05	Performance	Verify system response time for user actions	1. Navigate across multiple pages (Home, Events, Simulation) 2. Perform search and load actions	System responds within acceptable time thresholds
TC06	AI Insights	Verify user can ask portfolio risk questions	1. Navigate to Portfolio Agent page 2. Enter a risk-related question (e.g., "What is my risk exposure?") 3. Submit query	System returns relevant and accurate risk explanation based on portfolio data
TC07	Navigation	Verify quick access buttons on homepage work correctly	1. Navigate to Home page 2. View quick access section 3. Click each shortcut	User navigates to respective feature pages with a single click



Sprint Summary: Sprint 0

Sprint	Task Name	Status
Sprint 0	Project Vision Definition	Done
Sprint 0	Problem Statement	Done
Sprint 0	Team Roles & Responsibilities	Done
Sprint 0	Personas Definition	Done
Sprint 0	Technology Stack Selection	Done
Sprint 0	GitHub Repository Setup	Done
Sprint 0	GitHub Wiki Documentation	Done
Sprint 0	Establish Weekly Meetings	Done
Sprint 0	Work on Deliverable 1	Done

Sprint Summary: Sprint 1 User Stories

☐ ▼ Team-8 Sprint 1 24 Feb – 10 Mar (21 work items)			51 10 27	Complete sprint	...
Front-end Design, Create MVP Prototype, Create System Diagrams, Create Acceptance Criteria, Create Test Cases, Create Mock-up Web Design Page, Create User Stories or Tasks, Work on Deliverable 2, Working API, Application ...					
🔖 T8-41	As a potential investor, I want to understand RippleWealth's proactive risk intelligence approach to help me decide w...	TODO ▼	2	👤	
🔖 T8-44	As an investor, I want to view a projected portfolio value chart after simulation so that I can visually understand the ...	TODO ▼	3	👤	
🔖 T8-47	As an investor, I want to see portfolio risk metrics so that I can understand potential exposure to market volatility.	TODO ▼	3	👤	
🔖 T8-50	As a user, I want to view event details such as location, industry impact, and severity so that I can better understand...	TODO ▼	8	👤	
🔖 T8-51	As an investor, I want to run an event simulation so that I can understand the potential impact of a global event on m...	TODO ▼	3	👤	
🔖 T8-52	As an investor, I want to compare my current portfolio with recommended adjustments so that I can rebalance my in...	TODO ▼	3	👤	
🔖 T8-62	As an investor, I want to receive investment suggestions based on global events so that I can reduce portfolio risk.	TODO ▼	8	👤	
🔖 T8-63	As an investor, I want to understand Value at Risk so that I can estimate the maximum likely loss in normal market co...	TODO ▼	8	👤	
🔖 T8-64	As a portfolio manager, I want to receive automated rebalancing suggestions, so that I can reduce my exposure to hi...	TODO ▼	5	👤	
🔖 T8-65	As a user, I want to browse and filter real-world disruption events, so that I can stay informed about specific crises ...	TODO ▼	3	👤	
🔖 T8-66	As an investor, I want early investment advice so that I can move my money to safer assets when the market becom...	TODO ▼	5	👤	

Sprint Summary: Sprint 1 Tasks and Burndown Chart

Team-8 Sprint 1

24 Feb – 10 Mar (23 work items)

0

15

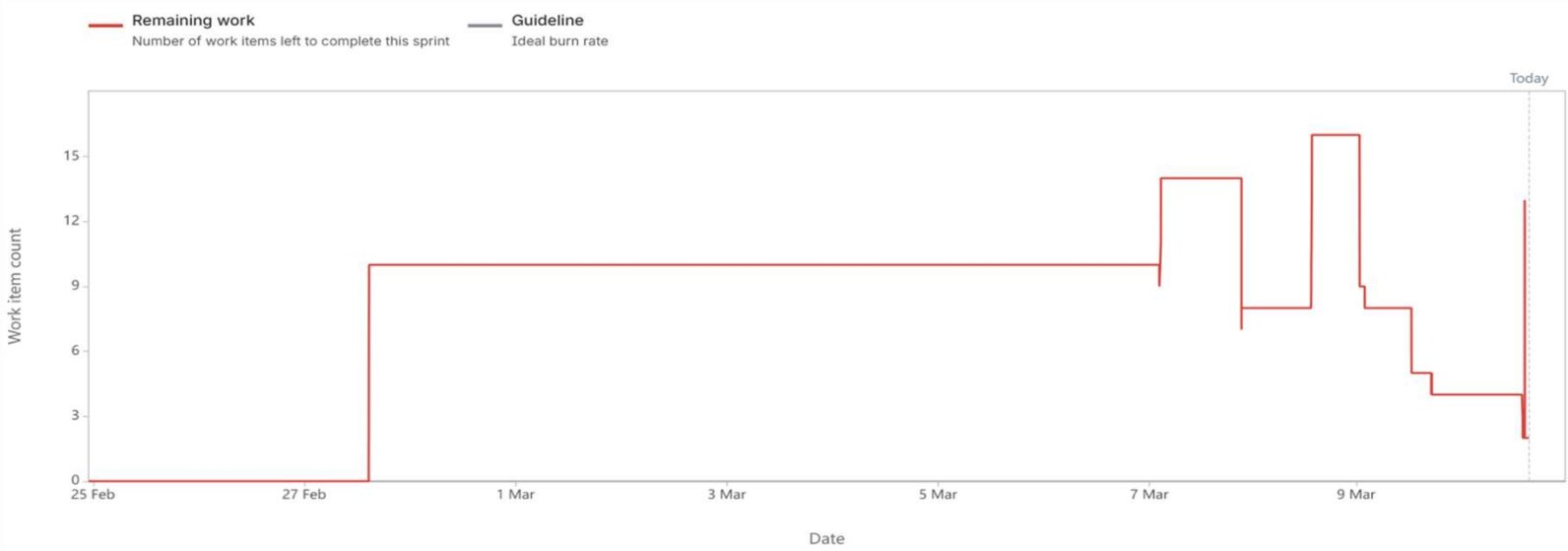
77

Complete sprint

..

Front-end Design, Create MVP Prototype, Create System Diagrams, Create Acceptance Criteria, Create Test Cases, Create Mock-up Web Design Page, Create User Stories or Tasks, Work on Deliverable 2, Working API, Applicati...

T8-10	Front-end Design	IN PROGRESS	5	DS
T8-11	Create MVP Prototype	DONE	8	AM
T8-12	Create System Diagrams	DONE	5	SK
T8-13	Create Acceptance Criteria	DONE	2	DS
T8-14	Create Test Cases	DONE	2	DS
T8-15	Create Mock-up Web Design Page	DONE	3	JP
T8-16	Create User Stories or Tasks	DONE	2	AM
T8-17	Work on Deliverable 2	DONE	2	SK
T8-18	Working API	IN PROGRESS	5	SK
T8-19	Application Demo	DONE	3	JP



Sprint Summary: Sprint 2 User Stories

Team-8 Sprint 2

11 Mar – 7 Apr (25 work items)

65

0

0

Complete sprint

...

T8-48	As an investor, I want to see my portfolio's Value-at-Risk (VaR), so that I understand the potential losses I could face during market disruptions.	TO DO	8	
T8-54	As an investor, I want the system to calculate potential portfolio losses using Monte Carlo simulations, so that I can understand the range of possible fin...	TO DO	8	
T8-55	As an investor, I want the platform to generate explanations for its risk predictions, so that I can understand the reasoning behind the AI insights.	TO DO	8	
T8-56	As an investor, I want to upload my portfolio holdings, so that the system can analyze how global events may affect my investments.	TO DO	3	
T8-58	As an investor, I want to simulate hypothetical events like natural disasters or geopolitical crises, so that I can understand how such events might affect ...	TO DO	8	
T8-59	As a developer, I want to implement Portfolio & Risk API Endpoints, so that the frontend can securely request and receive risk intelligence data.	TO DO	5	
T8-60	As an investor, I want the platform to generate natural language explanations for its risk predictions, so that I can understand the underlying reasoning b...	TO DO	8	
T8-61	As a user, I want to view a centralized risk dashboard with a sector exposure heatmap, so that I can visually identify if my investments are too concentra...	TO DO	5	
T8-45	As a portfolio manager, I want to filter events by region, severity or industry so that I can focus only on disruptions relevant to my investments	TO DO	3	
T8-43	As as registered user, I want to recover my password, so that I can regain access to my account without losing data.	TO DO	2	
T8-42	As a returning user, I want to click Login from the landing page, so that I can securely access my dashboard.	TO DO	2	
T8-67	As a user, I want the frontend to fetch live data from backend API's, so that all the pages display dynamic and updated information	TO DO	5	
T8-68	As a user, I want quick access buttons on the home page, so that I can move directly to the feature I need.	TO DO	-	
T8-69	As a user, I want loading and error messages to appear when backend data is being fetched or fails, so that I understand the current system state.	TO DO	-	

Sprint Summary: Sprint 2 Tasks and Burndown Chart

Team-8 Sprint 2

11 Mar – 7 Apr (25 work items)

0

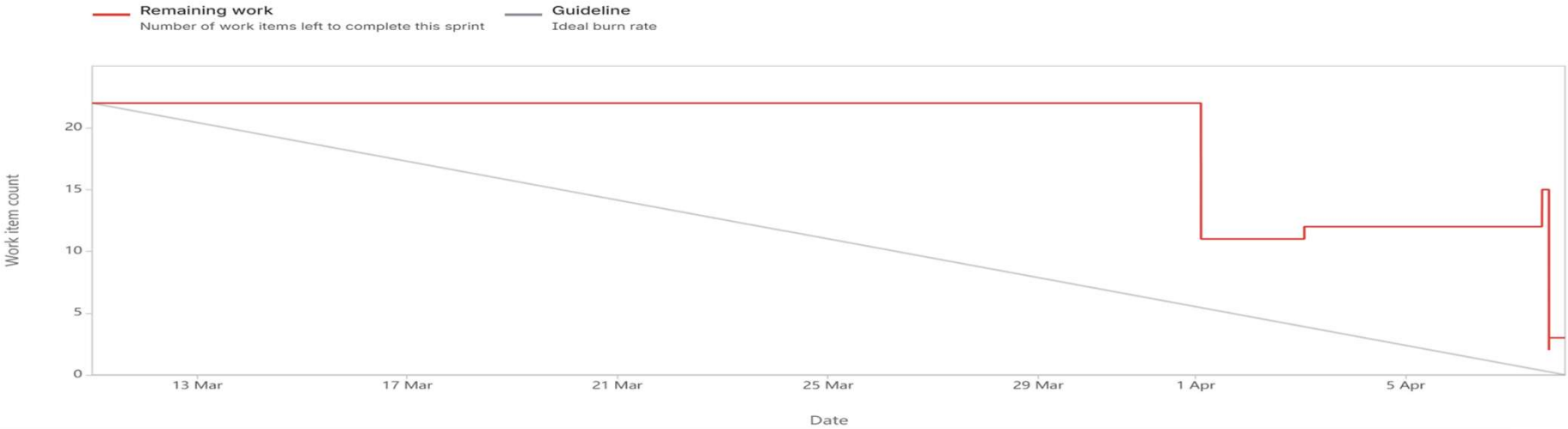
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134

Complete sprint

	 T8-20	Setup FastAPI Project & Database Models	DONE ✓	3	
	 T8-21	Implement Portfolio & Risk API Endpoints	DONE ✓	5	
	 T8-22	Causal Event Modeling	DONE ✓	8	
	 T8-23	Knowledge Graph Visualization Structure	DONE ✓	8	
	 T8-24	LLM Reasoning Logic	DONE ✓	8	
	 T8-25	Event Impact Simulation	DONE ✓	12	
	 T8-26	AI-Backend Integration	DONE ✓	5	
	 T8-27	Draft Technical Paper	DONE ✓	5	
	 T8-28	Develop Frontend	DONE ✓	8	
	 T8-29	Develop Backend	DONE ✓	5	
	 T8-30	Work on Deliverable 3	DONE ✓	3	

Date - 11 March 2026 - 7 April 2026



Sprint Backlog: Sprint 3

☐ Team-8 Sprint 3 8 Apr – 5 May (14 work items)	26038	Complete sprint	...
T8-31 Finish Development of Frontend	DONE ▾	8	DS
T8-32 Finish Development of Backend	DONE ▾	5	AM
T8-33 Frontend-Backend Integration Working	DONE ▾	5	SK
T8-34 System Testing	DONE ▾	5	JP
T8-35 Finish Project Demo	DONE ▾	5	JP
T8-36 Finish Technical Paper	DONE ▾	5	SK
T8-37 Work on Deliverable 4	DONE ▾	5	DS

Stories Completed - Sprint 3

☐ Team-8 Sprint 3 8 Apr – 5 May (14 work items)	0559	Complete sprint	...
 T8-70 As an investor, I want to complete the full workflow from portfolio uplo...	DONE ▾	8	
 T8-71 As a user, I want secure session handling (login/logout/session timeout...	DONE ▾	5	
 T8-72 As a user, I want the application to work across different browsers, so t...	DONE ▾	3	
 T8-73 As an investor, I want event data to refresh dynamically, so that I stay ...	DONE ▾	3	
 T8-74 As a user, I want fast response times, so that I can interact efficiently.	DONE ▾	2	
 T8-75 As an investor, I want to search for specific stocks in my uploaded port...	DONE ▾	3	
 T8-76 As a user, I want quick access buttons on the home page for direct nav...	DONE ▾	2	

Metrics - Status Overview as of Sprint 3

Spaces

RippleWealth  



[Summary](#)
[Timeline](#)
[Backlog](#)
[Board](#)
[Calendar](#)
[List](#)
[Forms](#)
[Development](#)
[Code](#)
[More](#)
[4](#)
[+](#)

Status overview

Get a snapshot of the status of your work items. [View all work items](#)



■ Done: 17

Metrics - Team's Historical Velocity (Average)

Team Historical Velocity (Average)

Sprint 1, Sprint 2, Sprint 3 and Average



Average is calculated as the mean of Completed Story Points for all sprints shown. The Average bar below represents the mean of Completed Story Points only $((83+137+64)/3 = 94.67)$.

Metrics - Sprint 3 Burndown Chart

Spaces / RippleWealth / Reports

Sprint burndown chart

> How to read this report



Sprint

Team-8 Sprint 3

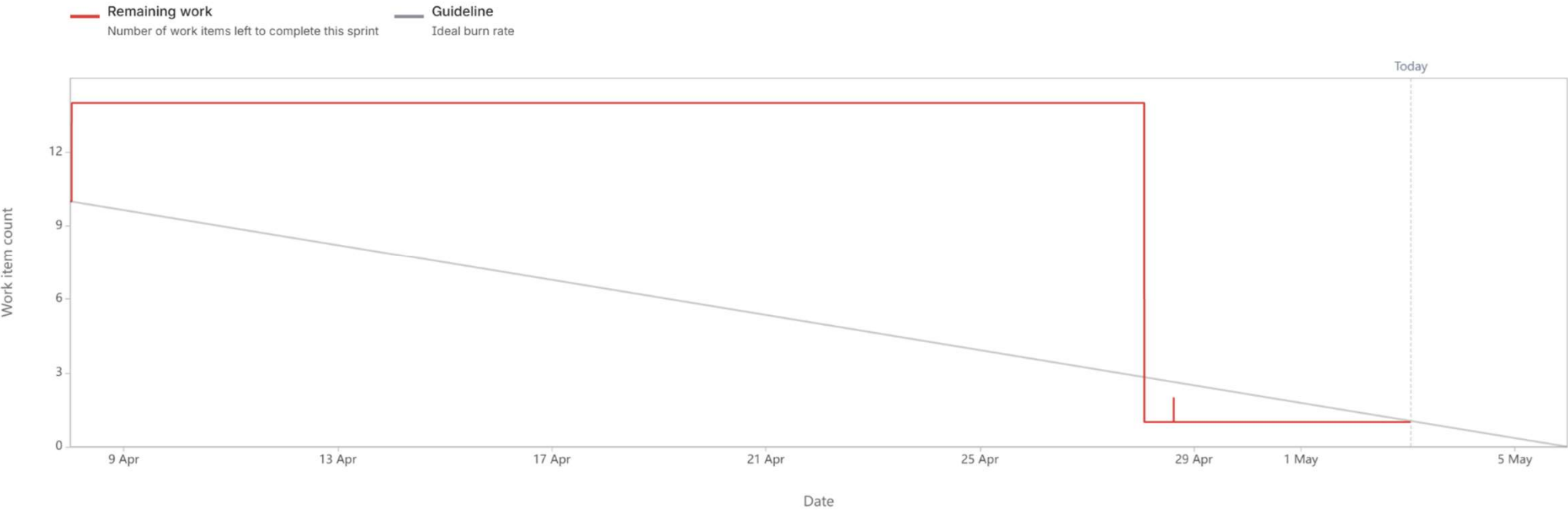
Estimation field

Work item count



Date - 8 April 2026 - 5 May 2026

Sprint goal - Finish Development of Frontend, Finish Development of Backend, Frontend-Backend Integration Working, System Testing, Finish Project Demo, Finish Technical Paper, Work on Deliverable 4



Metrics - Sprint 3 Ratio Status

Committed Story Points: 64
Completed Story Points: 64

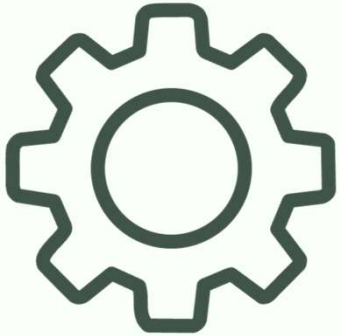
Completed/Committed
Ratio (Story Points) = $64/64$, which is 100.00%

Sprint 3 Story Points Completion (100.00%)



■ Completed

Retrospective



WHAT WENT WELL

- **Event Adjusted Simulation:**

Integrating live market events directly into scenario projections gave the application a differentiated capability.

- **Live Events Feed:**

The dual auto-scrolling marquee with category tagging and impact ratings shipped polished and on time, and the hover-to-pause interaction works smoothly.

- **Knowledge Sharing Across Roles**

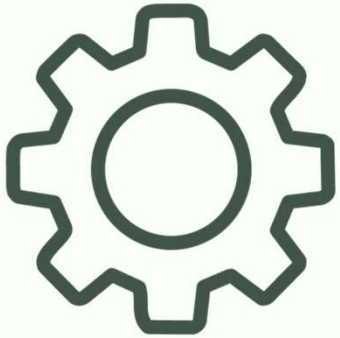
Frontend and backend members actively helped each other debug integration issues rather than working in silos.

- **AI Analysis End-to-End:**

The full pipeline from user holdings → yfinance price data → Sharpe, VaR, optimised weights → written AI explanation shipped completely and correctly within the sprint.



Retrospective



WHAT NEEDS IMPROVEMENT

- **Task Granularity and Estimation:**

Complex technical stories were taken on without being broken into smaller sub-tasks. This made it difficult to track progress mid-sprint.

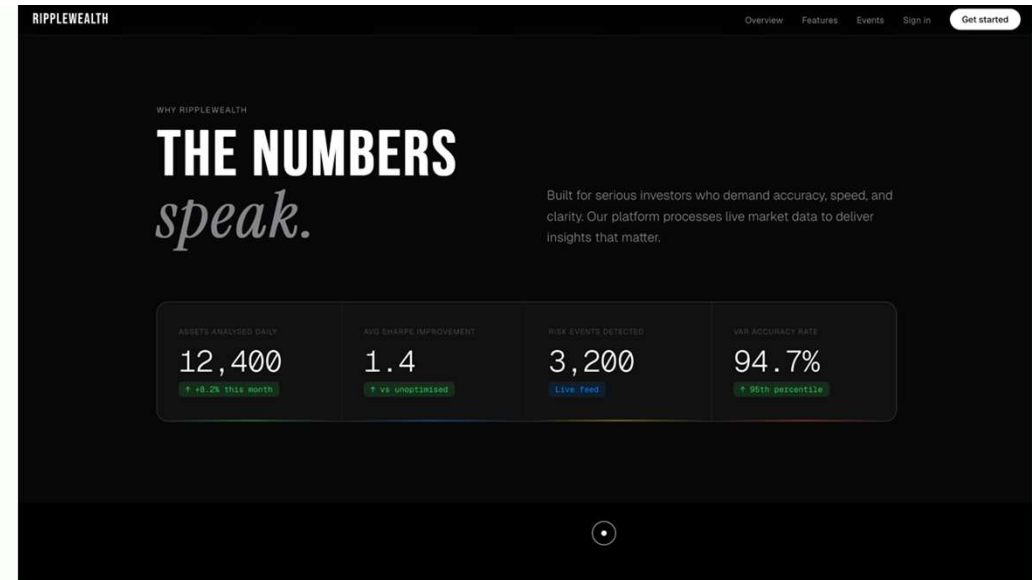
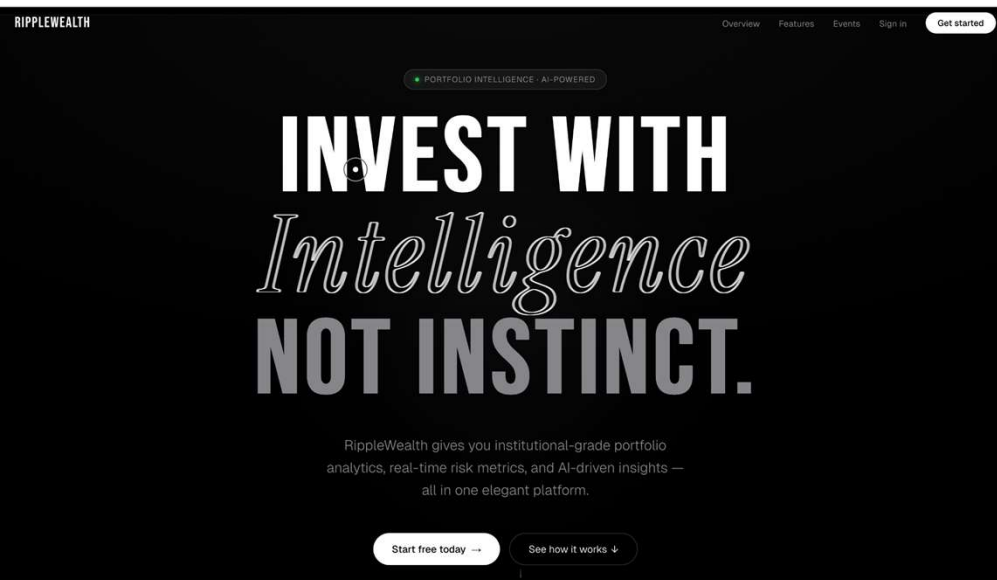
- **Spreading Integration Work Throughout the Sprint:**

Integration between the frontend and backend tended to happen toward the end of the sprint. Doing small integration checks more regularly throughout the sprint would make the final days feel much less pressured and give more time to polish.

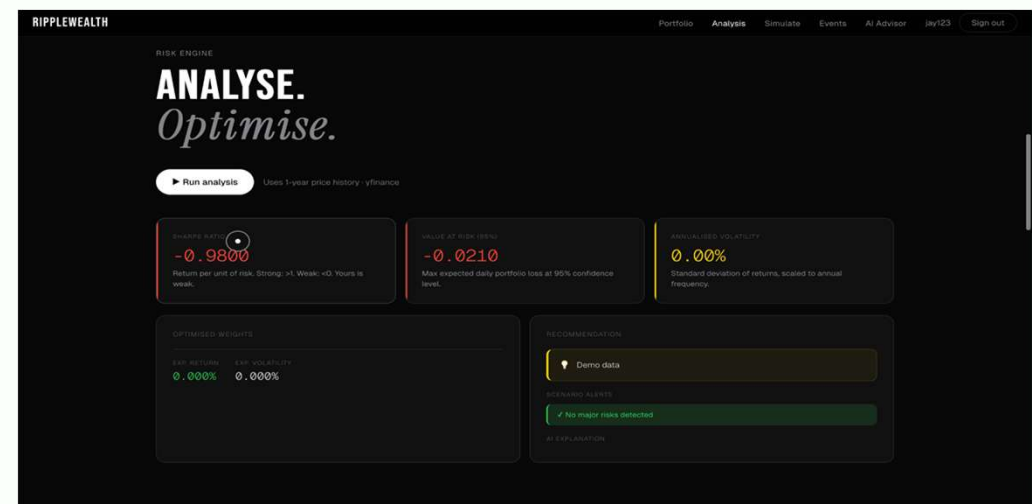
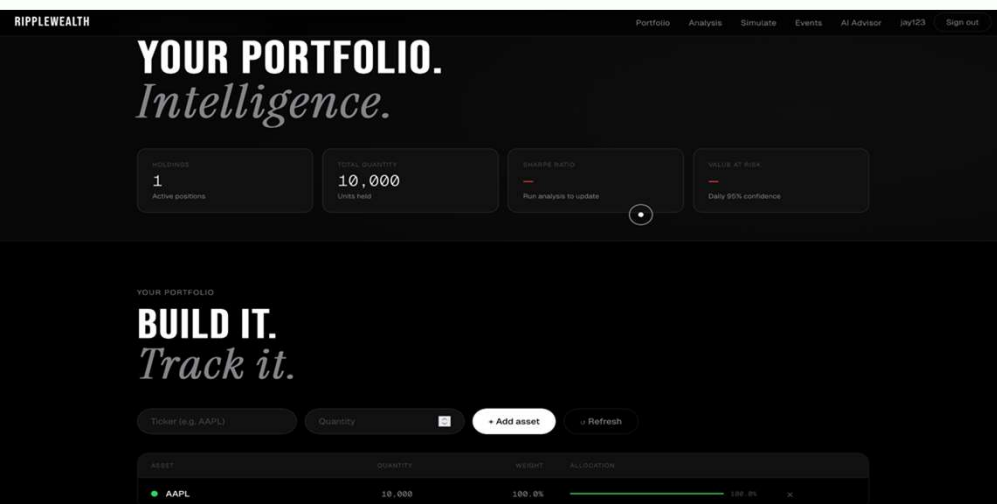


Project Demo - Sprint 3

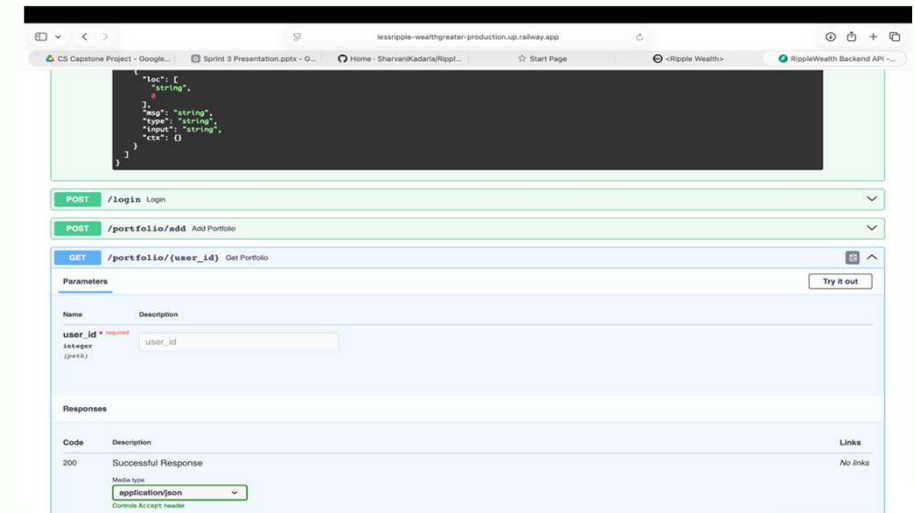
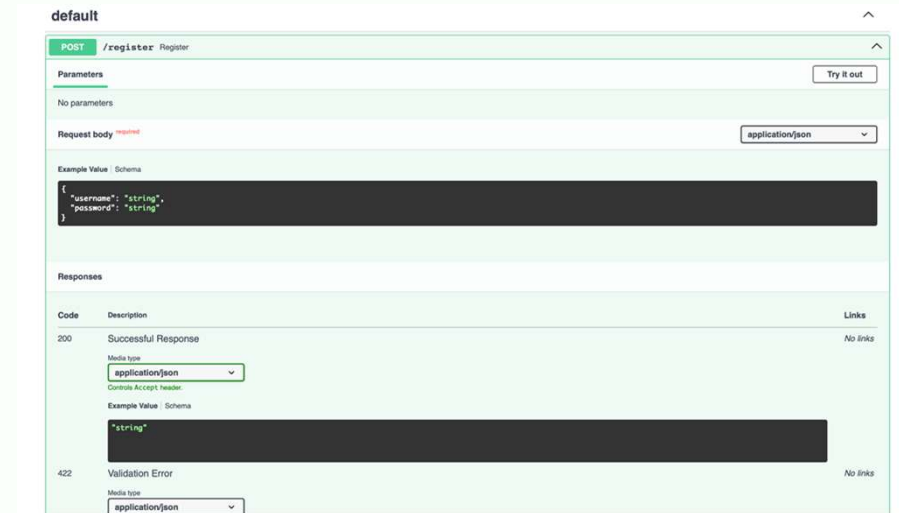




Application Screenshots



Working API Screenshots



Live Application Demo

GitHub Wiki Page Link



[SharvaniKadarla/RippleWealth Wiki Page \(github.com\)](#)

View on GitHub

Technical Paper Link



[RIPPLEWEALTH TECHNICAL PAPER \(GITHUB.COM\)](#)

View on GitHub

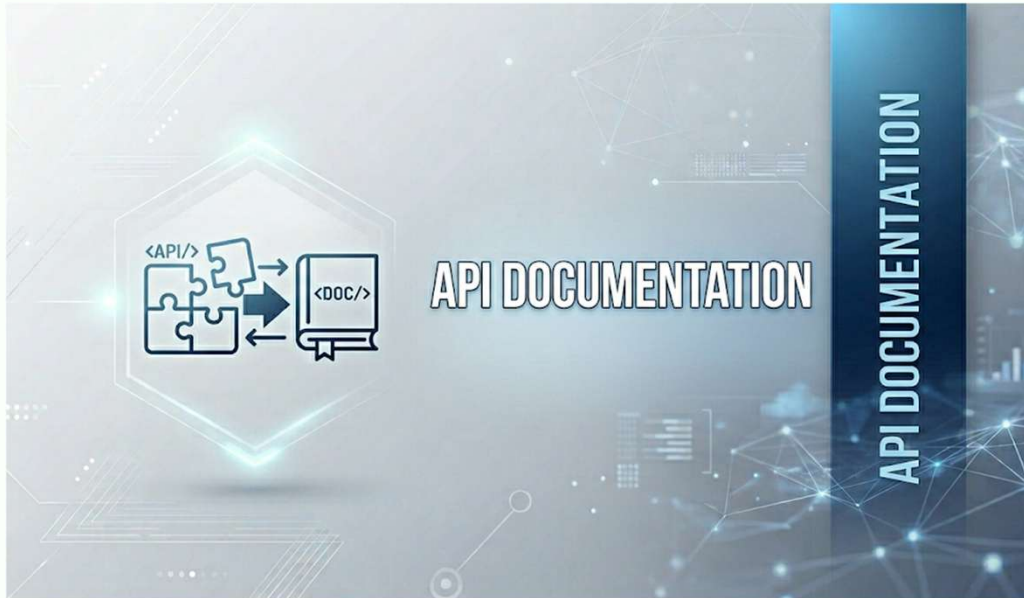
Deployment Manual Link



[RIPPLEWEALTH DEPLOYMENT MANUAL \(GITHUB.COM\)](#)

View on GitHub

API Documentation Link



[RIPPLEWEALTH API DOCUMENTATION \(GITHUB.COM\)](https://github.com/ripplewealth/ripplewealth-api-docs)

View on GitHub

THANK YOU